ElectViz Election Data Visualization for Media_Batch

Project Report

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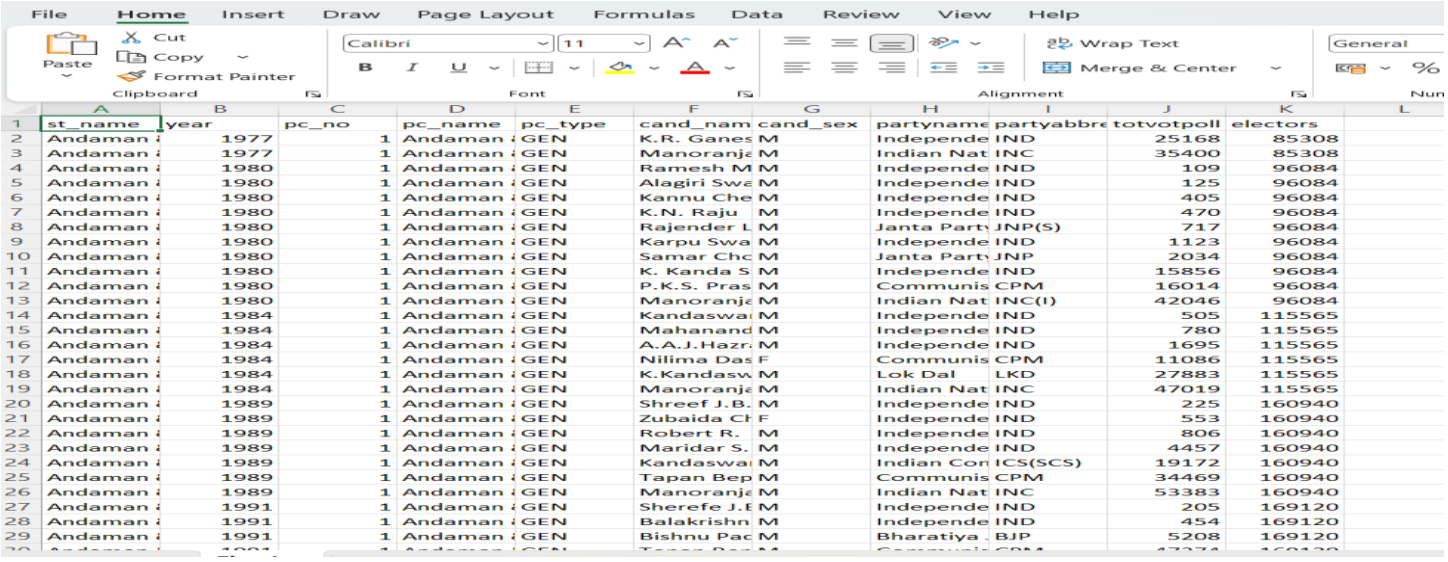
1.Project Introduction:

The **ElectViz project** aims to transform complex election datasets into clear and meaningful visual insights. Election **data contains large amounts of information such as candidate details, constituencies, political parties, votes received, and voter turnout across multiple years, making it difficult to analyze in raw tabular form.** To address this, the project converts raw election data into **interactive dashboards using Microsoft Power BI**. These dashboards allow users to explore election trends, compare results across different years, and analyze party and candidate performance. Using features such as filtering, drill-down analysis, and interactive charts, users can easily understand patterns in the data. The project analyzes **Indian General Election data from 1951 to 2019**, providing insights into voter turnout trends, party dominance, and overall election dynamics.

2.Problem Statement :

Election data is large and complex, containing details about candidates, political parties, constituencies, votes received, voter turnout, and election years, which makes it difficult to analyze in raw form. Most election data is stored in tabular or spreadsheet formats that only contain numbers and text, making it hard to identify patterns such as party dominance or voter participation trends. Media organizations and political analysts require quick insights from election results, but manually analyzing large datasets using spreadsheets is time-consuming and inefficient. In addition, the lack of interactive analysis tools makes it difficult to filter and compare data across years, parties, states, or constituencies. Raw election datasets may also contain duplicate records, missing values, and inconsistent data, which must be cleaned before analysis. Furthermore, identifying trends such as voter turnout growth, close election contests, and re-election patterns requires complex calculations. To address these challenges, there is a need for a system that can convert raw election data into clear and interactive visual insights. The ElectViz project solves this problem by cleaning the dataset and presenting the analysis through interactive Power BI dashboards, enabling easier understanding of election trends and results.

3. Dataset Used:



st_name	year	pc_no	pc_name	pc_type	cand_nam	cand_sex	partyname	partyabbre	totvotpoll	electors
Andaman	1977	1	Andaman	GEN	K.R. Ganes	M	Independe	IND	25168	85308
Andaman	1977	1	Andaman	GEN	Manoranja	M	Indian Nat	INC	35400	85308
Andaman	1980	1	Andaman	GEN	Ramesh M	M	Independe	IND	109	96084
Andaman	1980	1	Andaman	GEN	Alagiri Swa	M	Independe	IND	125	96084
Andaman	1980	1	Andaman	GEN	Kannu Che	M	Independe	IND	405	96084
Andaman	1980	1	Andaman	GEN	K.N. Raju	M	Independe	IND	470	96084
Andaman	1980	1	Andaman	GEN	Rajender L	M	Janta Party	JNP(S)	717	96084
Andaman	1980	1	Andaman	GEN	Karpu Swa	M	Independe	IND	1123	96084
Andaman	1980	1	Andaman	GEN	Samar Cho	M	Janta Party	JNP	2034	96084
Andaman	1980	1	Andaman	GEN	K. Kanda S	M	Independe	IND	15856	96084
Andaman	1980	1	Andaman	GEN	P.K.S. Pras	M	Communis	CPM	16014	96084
Andaman	1980	1	Andaman	GEN	Manoranja	M	Indian Nat	INC(I)	42046	96084
Andaman	1984	1	Andaman	GEN	Kandaswa	M	Independe	IND	505	115565
Andaman	1984	1	Andaman	GEN	Mahanand	M	Independe	IND	780	115565
Andaman	1984	1	Andaman	GEN	A.A.J.Hazr	M	Independe	IND	1695	115565
Andaman	1984	1	Andaman	GEN	Nilima Das	F	Communis	CPM	11086	115565
Andaman	1984	1	Andaman	GEN	K.Kandasw	M	Lok Dal	LKD	27883	115565
Andaman	1984	1	Andaman	GEN	Manoranja	M	Indian Nat	INC	47019	115565
Andaman	1989	1	Andaman	GEN	Shreef J.B.	M	Independe	IND	225	160940
Andaman	1989	1	Andaman	GEN	Zubaida Ch	F	Independe	IND	553	160940
Andaman	1989	1	Andaman	GEN	Robert R.	M	Independe	IND	806	160940
Andaman	1989	1	Andaman	GEN	Maridar S.	M	Independe	IND	4457	160940
Andaman	1989	1	Andaman	GEN	Kandaswa	M	Indian Con	ICS(SCS)	19172	160940
Andaman	1989	1	Andaman	GEN	Tapan Bep	M	Communis	CPM	34469	160940
Andaman	1989	1	Andaman	GEN	Manoranja	M	Indian Nat	INC	53383	160940
Andaman	1991	1	Andaman	GEN	Sherefe J.E	M	Independe	IND	205	169120
Andaman	1991	1	Andaman	GEN	Balakrishn	M	Independe	IND	454	169120
Andaman	1991	1	Andaman	GEN	Bishnu Pac	M	Bharatiya	BJP	5208	169120

st_name: state name

year: from 1977-2019

pc_no : Unique numerical identifier for each parliamentary constituency.

pc_name : Name of the parliamentary constituency.

cand_name : Name of the candidate who contested the election.

pc_type: Indicates the **type of parliamentary constituency based on reservation status**, such as General, SC (Scheduled Caste), or ST (Scheduled Tribe).

cand_sex: Gender of the candidate (Male or Female).

partyname: Political party to which the candidate belongs.

partyabbreviaition: Political party abbreviation

totpoll – Total number of votes received by the candidate.

electors – Total number of registered voters in the constituency.

4. Dataset Cleaning Process:

1. Removing Duplicate Records

- The first step was to identify and remove **duplicate rows** in the dataset.
- Duplicate records can occur due to data entry errors or repeated data imports.
- To detect duplicates, the following key columns were checked:
    - **Election Year**
    - **State**
    - **Constituency Number (pc_no)**
    - **Candidate Name**
- If multiple rows contained the same values in these columns, they were treated as duplicates.
- Only one valid record was kept while the repeated entries were removed.

Purpose:
Removing duplicate records ensures that each candidate appears **only once per constituency per election year**, which prevents incorrect vote totals and inaccurate analysis.

2. Handling Missing Values in Vote Columns

- The dataset contained a column representing **total votes received by candidates (totpoll)**.
- Some rows had **blank or null values** in this column.
- If the candidate existed but the vote count was missing, the null value was replaced with **0**.
- Rows where both **candidate name and vote values were blank** were completely removed from the dataset.

Purpose:
Handling missing vote values ensures that calculations such as **total votes, vote share, and turnout percentage** are accurate and do not produce errors during analysis.

3. Cleaning Elector and Turnout Data

- The dataset also included a column representing **total registered electors** for each constituency.
- Some records had **null or missing values** in this column.
- These rows were removed because turnout calculations require valid elector data.

Purpose:
Turnout percentage is calculated using the formula:

$$\text{Turnout} = \frac{\text{Total Votes}}{\text{Total Electors}}$$

If elector values are missing, the calculation may produce incorrect results or errors. Removing such rows ensures reliable turnout analysis.

4. Standardizing Gender Information

- The dataset contained a column representing the **gender of candidates (cand_sex)**.
- Some rows had missing or blank gender values.
- These missing values were replaced with the category **"Other"**.

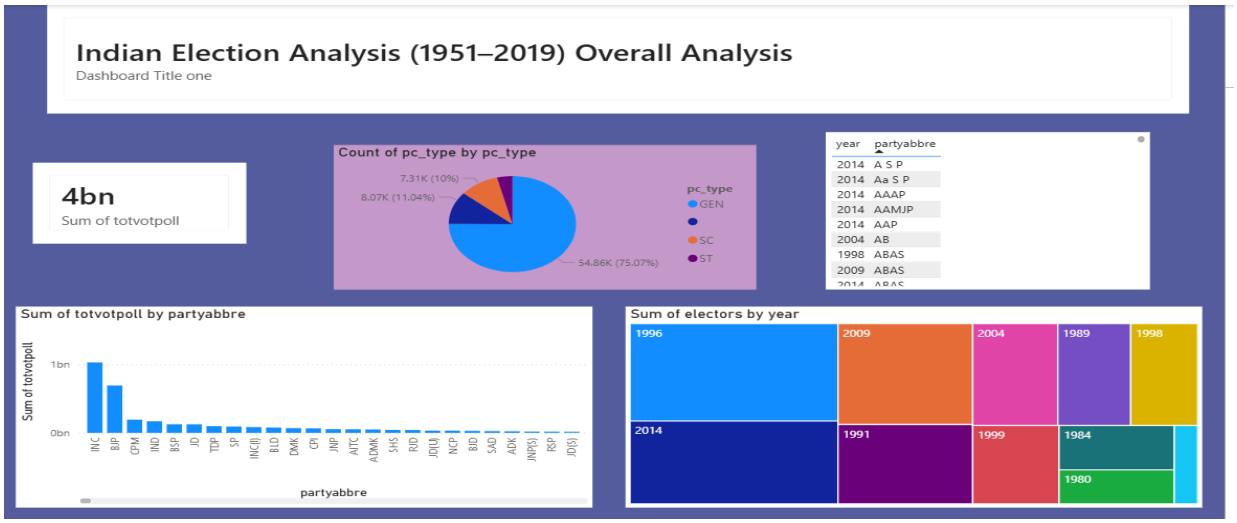
Purpose:
Standardizing gender values ensures that demographic analysis (such as male vs female candidate participation) can be performed without missing data affecting the results.

5. Validating Candidate and Constituency Data

- Candidate names and constituency names were checked to ensure they were **not empty or incorrectly formatted**.
- Rows where essential fields such as **candidate name or constituency name were missing** were removed.

Purpose:
This ensures that each record represents a **valid candidate contesting in a specific constituency**, which is necessary for accurate election analysis.

Task 1 : Clean the dataset and load in power bi and prepare an overall analysis dashboard.



- **Overview of Indian General Elections (1951–2019)**
  This dashboard provides a summary of Indian General Election data from 1951 to 2019, helping users understand overall election patterns and trends.

- **Total Votes Card Highlights Cumulative Votes Cast Across Years**
  The total votes card shows the total number of votes cast across all election years, giving a quick overview of overall voter participation.

- **Constituency Chart Shows Majority of Seats are General Category**
  The constituency chart displays the distribution of constituencies by category (General, SC, ST), showing that most seats belong to the General category.

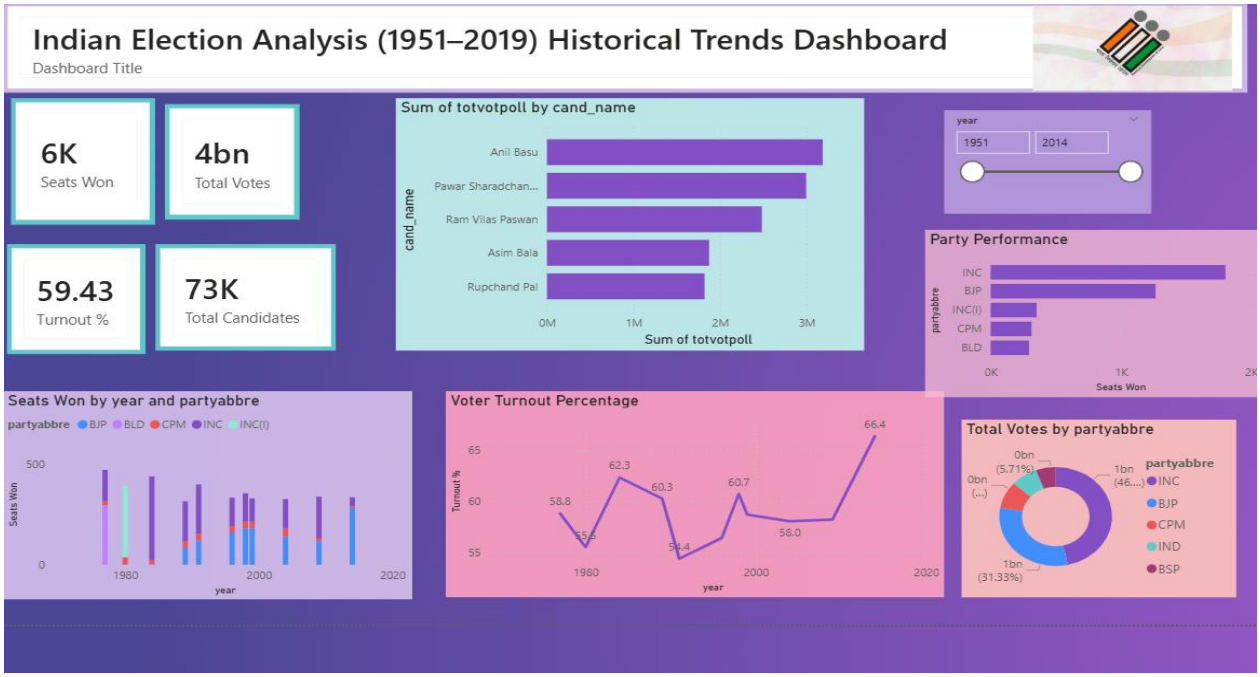
- **Party-Wise Chart Indicates INC and BJP Receive the Highest Votes**
  The party-wise chart compares votes received by different parties and shows that INC and BJP receive the highest vote share.

- **Treemap Shows Steady Growth in Registered Voters Over Time**
  The treemap visualization highlights the increase in registered voters across election years.

- **Year and Party Slicers Enable Interactive Filtering**
  Slicers allow users to filter the dashboard by year or party for better analysis.

- **Insight** :
  The dashboard highlights long-term election trends, increasing voter participation, and dominance of major political parties.

Task 2 : Prepare Historical Trends Dashboard



- **Shows Historical Trends of Indian General Elections (1951–2019)**
  This dashboard focuses on analyzing election patterns over time, helping users understand how election results and voter behavior have changed across different years.

- **KPI Cards Summarize Seats Won, Total Votes, Turnout %, and Candidates**
  The KPI cards provide a quick overview of key election metrics such as total seats won, number of votes cast, voter turnout percentage, and total candidates participating.

- **Candidate Votes Chart Highlights Top Candidates by Total Votes**
  This chart displays candidates who received the highest number of votes, helping identify the most popular or influential candidates.

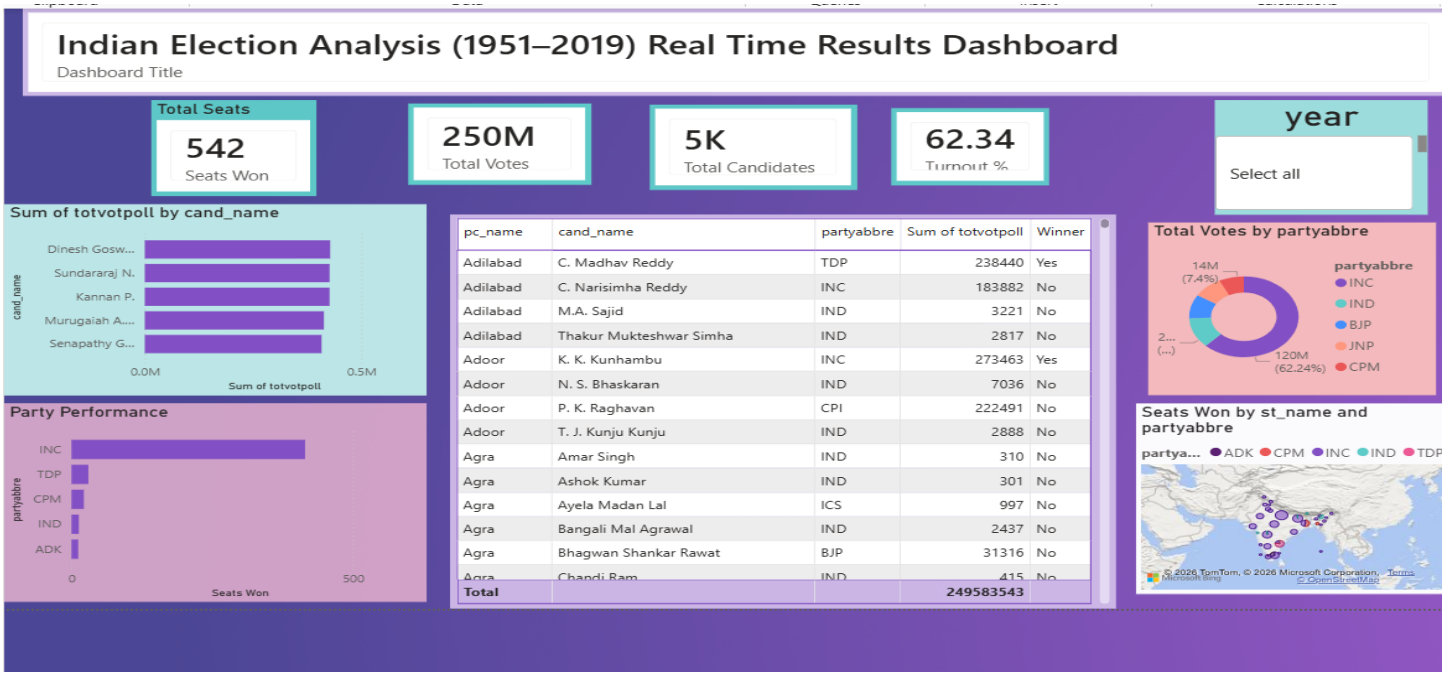
- **Year Slicer Enables Filtering and Comparison Across Election Years**
  The year slicer allows users to select specific election years and compare results across different time periods.

- **Party Performance Chart Shows Seats Won by Major Parties (INC, BJP Leading)**
  This chart compares the number of seats won by different political parties and highlights the leading parties such as INC and BJP.

- **Stacked Column Chart Shows Party Performance Changes Across Years**
  The stacked column chart illustrates how the performance of political parties has changed over multiple election years.

- **<u>Insight</u>**
  This dashboard helps reveal long-term party dominance, candidate popularity, and changes in political power over time.

Task 3 : Prepare Real Time Dashboard



pc_name	cand_name	partyabbre	Sum of totvotpoll	Winner
Adilabad	C. Madhav Reddy	TDP	238440	Yes
Adilabad	C. Narisimha Reddy	INC	183882	No
Adilabad	M.A. Sajid	IND	3221	No
Adilabad	Thakur Mukteshwar Simha	IND	2817	No
Adoor	K. K. Kunhambu	INC	273463	Yes
Adoor	N. S. Bhaskaran	IND	7036	No
Adoor	P. K. Raghavan	CPI	222491	No
Adoor	T. J. Kunju Kunju	IND	2888	No
Agra	Amar Singh	IND	310	No
Agra	Ashok Kumar	IND	301	No
Agra	Ayela Madan Lal	ICS	997	No
Agra	Bangali Mal Agrawal	IND	2437	No
Agra	Bhagwan Shankar Rawat	BJP	31316	No
Agra	Chandi Ram	IND	415	No
Total			249583543	

- **Shows Year-Specific Election Results in a Detailed Interactive View**
  This dashboard focuses on analyzing election results for a particular year, allowing users to view detailed information about that specific election.

- **KPI Cards Summarize Seats Won, Total Votes, Candidates, and Turnout %**
  The KPI cards provide a quick overview of important metrics such as the total number of seats won, total votes cast, number of candidates, and voter turnout percentage.

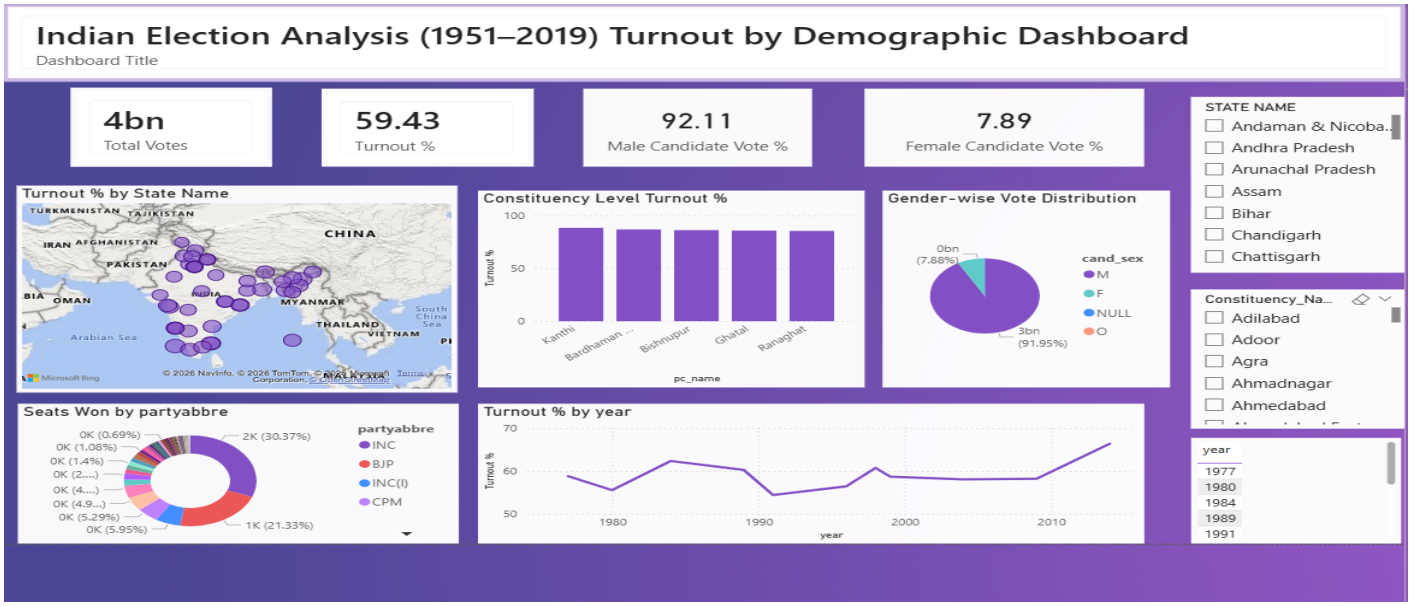
- **Year Slicer Filters the Dashboard for a Selected Election Year**
  The year slicer allows users to select a specific election year, and all visuals in the dashboard update accordingly.

- **Candidate Votes Chart Highlights Top Candidates in That Year**
  This chart displays candidates who received the highest number of votes in the selected election year.

- **Table Shows Constituency-Level Details**
  The table provides detailed information for each constituency, including the constituency name, candidate name, political party, total votes received, and the winning candidate.

- **<u>Insight</u>**
  This dashboard helps users perform detailed analysis of election results and evaluate candidate performance for a specific election year.

Task 4 : Prepare Turnout by Demographics Dashboard:



Turnout by Demographics Dashboard – Explanation (Short)

- **KPI Cards Summarize Total Votes, Turnout %, and Gender-Wise Vote Share**
  The KPI cards provide a quick summary of important metrics such as total votes cast, overall voter turnout percentage, and vote share based on candidate gender.

- **Map Shows Voter Turnout Percentage by State**
  The map visualization displays voter turnout across different states, helping identify regions with higher or lower voter participation.

- **Bar Chart Compares Turnout Across Constituencies**
  This chart compares voter turnout percentages across different constituencies to highlight variations in participation levels.

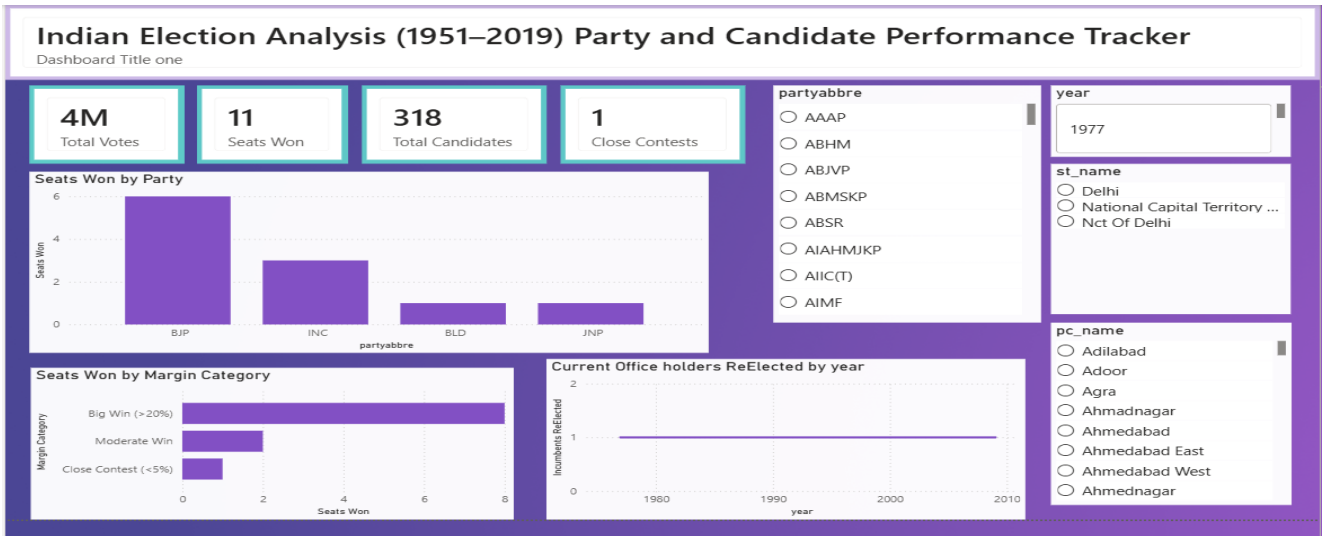
- **Pie Chart Shows Vote Share Between Male and Female Candidates**
  The pie chart illustrates the proportion of votes received by male and female candidates.

- **Donut Chart Shows Seats Won by Different Parties**
  The donut chart displays the distribution of seats won by various political parties.

- **Line Chart Shows Voter Turnout Trend Across Years**
  The line chart visualizes how voter turnout has changed across different election years.

- **<u>Insight</u>**
  This dashboard highlights voter participation patterns, regional differences in turnout, and gender representation in elections.

Task 5 : Prepare Party and Candidate Performance Tracker Dashboard



- **KPI Cards Summarize Total Votes, Seats Won, Candidates, and Close Contests**
  The KPI cards provide a quick overview of key election metrics such as total votes cast, number of seats won, total candidates participating, and the number of close contests.

- **Party Chart Shows Seats Won by Each Political Party**
  This chart displays the number of seats won by different political parties, helping identify the leading parties in the elections.

- **Margin Chart Categorizes Wins into Big, Moderate, and Close Contests**
  The margin chart classifies election victories based on the winning margin, showing whether the contest was a big win, moderate win, or a close contest.

- **Line Chart Shows Re-Election Trends of Incumbents Over Time**
  The line chart illustrates how often incumbent candidates were re-elected across different election years.

- **Filters Allow Analysis by Party, Year, State, and Constituency**
  Interactive filters enable users to analyze election results based on specific parties, years, states, or constituencies.

- **Insight**
  This dashboard highlights election competitiveness, party dominance, and trends in incumbent re-elections.

Conclusion

The **ElectViz project** successfully demonstrates how large and complex election datasets can be transformed into meaningful insights using data visualization techniques. By cleaning and preparing the dataset and developing interactive dashboards in **Microsoft Power BI**, the project enables users to analyze election data more efficiently. The dashboards provide valuable insights into voter turnout trends, political party performance, candidate popularity, and election competitiveness across multiple years.

Through interactive features such as filters, charts, and slicers, users can easily explore election results from different perspectives and compare trends across constituencies, states, and election years. This helps media organizations, analysts, and researchers better understand electoral patterns and political dynamics.

Overall, the project highlights the importance of **data visualization in simplifying complex datasets** and supporting data-driven analysis. The ElectViz dashboards provide a clear and interactive platform for studying election trends and gaining deeper insights into the functioning of democratic elections.